Linac Setup Manual

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1.0 PURPOSE

This document describes the principle and detailed actions required to adjust the linac to meet the operation requirements as the first stage of the injection system of the Canadian Light Source. This report provides detailed operational procedures of the CLS linac required to safely and efficiently produce a beam of electrons suitable for transport through the linac-to-booster line and injection into the CLS booster. The linac provides a pulse train of electrons from 2 ns to 136 ns in duration, with a nominal beam energy of 250 MeV and an average beam current of approximately 65 mA in the pulse train as measured at the end of ECS.

2.0 BACKGROUND

The linac is composed of many components that have to be adjusted for the operation. The report includes all aspects of linac operation up to the Energy Compression System (ECS).

The beam produced by the linac must have horizontal and vertical emittances not exceeding 0.3 mm-mrad, an energy spread of 2% or better and phase spread (at 2856 MHz) of about 12 degrees. The energy spread and phase requirements are necessary for subsequent reduction of the energy spread to 0.15% by the Energy Compression System (ECS). The small energy spread and transverse emittances are required for efficient transport to and injection into the CLS booster.

Due to the levels of radiation produced by the linac when it is in operation, the linac must be enclosed in a secure area (linac vault). The vault serves to exclude access to the linac environment while the linac is in operation and to shield the environment external to the secure area from the radiation that is produced. As well access to the secure area must be restricted until the residual levels of radiation have been measured and safe working conditions have been established.

For routine operation, when all equipment is ready to run and all interlocks have been made, the operator need only follow section 4. Section 3 describes the preparation for operation.

3.0 Preparation for Linac Operation

Before the electron beam can be injected into the linac, all equipment, including safety interlocks, vacuum, cooling and heat exchangers, pop-up monitors, power supplies, modulators, klystrons, slits, phase shifters, attenuators, timing and RF driver, etc., must be ready for running. Also, all six accelerating sections have to be conditioned. The RF power in the accelerating sections must reach the required levels. The operators should process them as follows:

3.1 Check and turn equipment on before linac beam running

The Linac cannot be running until whole system is ready for operation. It has to be checked and prepared before operation.

3.1.1 Vacuum system check

The vacuum pressure in the electron gun, prebuncher, accelerating sections, and waveguides leading from the klystrons to the accelerating sections must be kept lower than predetermined values before RF power is allowed to propagate through the waveguides to the RF cavities or before the electron gun is run. Loss of vacuum quality will result in the RF drive being disabled, and the sparking will trip the VSWR circuit and stop RF. Also, the gun is automatically disabled by any excessive vacuum pressure fluctuations.

- The vacuum system, including electron gun, waveguides and accelerating sections, must be checked before the linac setup to make sure all vacuum sections in the linac are good and all valves are open.
- The linac vacuum control screen can be accessed via the pull down menu under the CLS logo, under **Linac -> Vacuum Control**. The status of all vacuum sections will be displayed as either GOOD or BAD and the valves will be displayed as either OPEN or CLOSED.
- Vacuum section 1 is a part of the Linac vacuum system. Details of the linac ion pumps can be obtained under the **Details** section by selecting the **LINAC** button.
- IOP0001-01 to IOP0001-08 and their power supplies are located in the linac tunnel.
- IOP0001-09 to IOP0001-11 are located in the linac vault and service the long waveguide. Their power supplies are located in power supply room 0102.
- IOP1032-02 to IOP1032-12 are waveguide vacuum pumps located in room 1032 and the power supplies are located in the control room rack R1021.5-06 behind the main console.

- Also, six klystrons have their own vacuum ion pumps built on the klystrons. The ion pumps are located in the modulator room 1032. Their power supplies are located behind the main console in control room in rack R1021.5-06.

3.1.2 Interlock check

The interlock system consists of two control parts: the old 115V interlock system and the new 24V micro 84 programmable interlock controller.

Also, the system can be divided into many parts: vacuum interlocks; seven zone doors; gate interlocks; radiation interlocks; micro 84 controller interlocks; and other equipment safety interlocks.

- The personnel safety interlocks, i.e., seven zone interlocks must be made to ensure that these areas have been evacuated and to keep people from entering the zones.
- All green and red zone interlock lights are located in rack R1021.3-01 at the back left side of the console. All lights must be lit to indicate the zones are ready for operation.
- The green lights located in rack R1021.1-10 panel P1021.1-02 of the control room console must be lit. These lights indicate that all accelerating sections' vacuums, access doors, radiation monitors and micro 84 controller are OK. (Refer to LINAC-LTB1 Lockup Procedure, Ref. 5).
- All other interlocks must be made (Refer to Health Safety and Environment reports on the Access Control Interlock System for details).

3.1.3 Cooling system and Heat Exchanger system check

Temperature stability of the accelerating sections to within ± 0.1ºC is important for stable operation of the linac. Temperatures will be monitored and can be checked if the beam stability becomes a problem. The linac can be turned off at the operator's discretion, at which time the temperature can be adjusted by a qualified technician.

Large variations in temperature will result in the gun and RF drive both being disabled by the control system. Heat exchanger performance and water flow rates are also monitored for possible indications of problems with the cooling systems.

- The linac heat exchangers control screen can be accessed via either selecting the **Heat Exchangers** icon -> **Operator** / **Diagnostics** or selecting the **RunLINAC** icon **-> Heat Exchangers.** The status of a heat exchanger will be displayed after selecting a detail device.

- For the sections that are used, the temperatures are #1, #5, #6----45°C; #0, #2, #3, #4---40°C; #7--- 43°C (+/- 0.1°C).
- If the sections are not used, detuned temperatures are 55°C.
- Klystrons' temperature---30°C.
- All other cooling circuits, including **Low Energy**, **ECS**, and **LTB** are only used for cooling so no precise temperature control is required. (However, damaged motors or pumps could result in heat buildup that may lead to increased temperature. The pumps and drive motors need to be checked by appropriate personnel if the temperatures are either higher than 30 °C or lower than 26 °C).

3.1.4 Timing system check

The timing system coordinates the timing among the major systems of the CLS including the linac, booster and the storage ring. For the linac, an array of timing signals is used to set up the proper sequence of events and to optimize the performance.

Triggers from the timing system are used to adjust the timing of each of the transmitters to optimize the RF power from each klystron. Details of the timing are given in the sections that follow. The timing system produces triggers that can be adjusted over hundreds of microseconds with a resolution of better than 1 ns. Typically, delays of a few microseconds are used.

For proper phasing of each section, the RF power to each section must be controlled within one degree of phase at 2856 MHz (about 1 ps). This part of the timing is achieved with the phase shifters that are part of the transmitters described above.

- The linac timing control screen can be accessed via the pull down menu under the CLS logo. Select **Timing.** The status of the triggers (timing) will be displayed.
- Make sure that the trigger mode LINAC/LTB is selected. From this screen, the triggers for the **RF**, **Modulators**, and **Gun** can be accessed at the bottom of this screen. This screen also allows access to all of the delays associated with the modulators and gun operation. For linac setup, select the **RF**, **Modulators**, and **Gun** triggers.
- The normal repetition frequency should be set at 1.00 Hz (**MUST BE LESS THAN 3 Hz!**).

3.1.5 Pop-up monitors check

"Pop-up" monitors are required where the beam energy or the beam intensity is too low to produce a good transition radiation (TR) signal. Initially, pop-up monitors will be installed at locations along the linac. Pop-up monitors crudely measure both the size and position of the electron beam. Before running the electron beam, all pop-up monitors should be systematically checked using the following procedure:

- Press the switch buttons located in rack R1021.1-09, panel P1021.1-26 on the control room console.
- If these pop-up monitors work well, the targets should be clearly displayed on a screen.
- If there is no image or the image is dark, either the camera or the camera light may require service.

3.1.6 Power supplies check and turn on

Power supply failures may result in the mis-steering or defocusing of the beam in the linac. The beam may disappear.

- The power supply control screen can be accessed via selecting the **Magnets** icon **-> Power Supplies.** The status of the power supplies will be displayed.
- Selecting **On** or **Off** buttons, the power supplies can be switched on or off.
- Selecting **PSxxxx-xxx** buttons, the detailed channels information can be displayed. DO NOT SELECT the **<u>Reboot</u>** BUTTON**.**
- All linac related power supplies should be turned on before running the linac.
- To check which magnets the power supplies drive, refer to the **Linac Magnet Controls Computer System Design** Drawing No. ACCL/SE/CPT/0066103. One of these sheets will be located in the control room.
- Also, via selecting the **Linac & LTB Map** icon, then magnet's button, the magnet's detail information can be displayed.

3.1.7 Transmitters check and turn on

Each transmitter consists of a modulator and a klystron. The modulators determine the duration and amplitude of the high power RF signal supplied by each klystron. Each modulator supplies a high voltage pulse to a klystron for about 1-2.5 μs. The amplitude of the high voltage from the modulator together with the amplitude of the RF drive signal determine the amount of RF power delivered by each klystron to the accelerator sections. Optimization of the power delivered to each section is a key factor for stable operation. RF power is produced by each klystron when the RF drive and modulator high voltage are both present.

The klystrons provide the RF power that is supplied to the accelerating sections of the linac. The distribution of RF power is as follows:

Klystron 1	Prebuncher, Section 0, Section 1, ECS (Section 7)
Klystron 2	Section 2
Klystron 3	Section 3
Klystron 4	Section 4
Klystron 5	Section 5
Klystron 6	Section 6

Check the status of the transmitter and turn it on:

- The linac transmitter control screen can be accessed via selecting the **RunLINAC** icon **-> Modulators.** The status of the modulators will be displayed.
- The modulator high voltages should be set to a value of approximately 40 kV. If the voltage displayed on the modulator screen varies by more than 2.5 kV on any of the chosen modulators, you should check the number with a qualified technical person. (Sometimes after a PLC reboot, the appropriate value needs to be selected and then <Enter> pressed to activate the HV power supply).
- Selecting the numbered buttons, the transmitter's detailed status can be displayed on the pop-up windows. For operation, all information should be green.
- Any problem with the vacuum, temperature control, water flow, doors, over/under current, VSWR, or fan should be solved by authorized personnel. If the filament is off, follow the steps below.
- First, the operator should turn on the modulator power supply by using the toggle switches located near the top of the modulator control cabinets in room 1032. The toggle switches are located on panels P1032.1-01 for modulators 1 and 2, P1032.3-01 for modulators 3 and 4, and P1032.5-01 for modulators 5 and 6.
- Second, depress the six green **STANDBY** push buttons located on panels P1032.1-02, P1032.3-02, and P1032.5-02. Activating these buttons will allow the six klystron and six thyratron filaments to be turned on.
- After 20 minutes waiting for heating the filament and cathode, in the modulator control window, all status lights in the control window should change to green and the status right besides the numbered buttons should be **GOOD.**
- Select all six **YES** buttons under **Required**. This step turns one of the RF driver switches on, and will allow an RF signal to be sent to the klystrons (another series connected switch related to interlocks also should be made for sending RF signal to the RF drive klystron). Now, the red **DISABLE** lights will be displayed because the **HV** (high voltage) buttons have not yet been turned on.

- The high voltage supplies for each modulator can now be turned on. First, the rocker switch on each rack mounted power supply must be turned on. The 6 power supplies are located below the modulator control panels in room 1032. The power supplies are labeled PS1032.1-01, PS1032.1-02, PS1032.3-01, PS1032.3-02, PS1032.5-01, and PS1032.5-02.
- Next, the modulator high voltage can be turned on. This can be accomplished by one of three methods. 1) Normally, the modulator high voltage is turned on by selecting all **HV->ON** buttons in the modulator control window. 2) Alternately, the HV can be turned on using the green push buttons located in rack R1021.1-09 on panel P1021.1-24 on the control room console. 3) The modulator high voltage can also be turned on in the modulator room using the 6 green push buttons labeled **HV ON** on panels P1032.1-02, P1032.3-02, and P1032.5-02. Now, the Required RF & Gun should display *green* **ENABLE**, on the control window.
- If all interlocks are good, the system is ready to have RF power sent to the accelerating sections. If the sections have been conditioned, it is a good time to turn on the RF to make sure that all systems are functioning. Turning on the RF can be accomplished by depressing the red push button located on panel P1021.1-22 of the main control room console.
- To check that RF power is being applied to the sections, The RF pulses can be displayed on an oscilloscope. This is accomplished by connecting the oscilloscope to the BNC connectors labeled **Fwd.**, **Rev.**, or **Load** located on panel P1021.1-12 on the main control room console. The RF forward power, reverse power, and power transmitted through the section to the load can be measured by using the 7 position rotary switches on the same panel.
- If RF power is not observed, the transmitter high voltages and currents pulses can be displayed on an oscilloscope by connecting a cable to the BNC connectors labeled **Volt** or **Curr.** on panel P1021.1-13. The pulses for each section can be selected by using the 7 position rotary switches located on the same panel. If no RF power is observed, you should contact a qualified technical person to investigate.

3.1.8 Slits check and calibration

All Linac & LTB slits must be calibrated and left in the open position.

- Operation of the slits can be accessed on the Linux (control) computers operating the general computer interface, selectinging the **Run Linac** icon.
- Selecting the **Linac Slits** button brings up a control screen where all of the Linac and LTB slits can be controlled.

- A green light will be present if the slits have been calibrated. If not, the **Start** button can be pressed to do the calibration. If the light is still red after the calibration, do the calibration again. (Sometimes, the calibration is unsuccessful.)
- After calibration, all slits should be opened to 20 mm in the **gap** edit box, and, in general, the **center** should be set to 0.00 mm.

3.1.9 Beam targeting setup

The targeting option must be set to enable the proper interlocks for running the linac to the desired end location.

- The targeting options are accessed via the pull-down menu located under the CLS logo, under **Machine Setup**.
- The orange **Interlocks =>Linac Interlocks** button can be used to check the status of the beam targeting interlocks.
- If the green dot is beside the **NO Target** button, there will be no beam ejected out from the electron gun.
- Selecting an appropriate button (egs. **Booster**)**,** the green dot will be relocated beside the button. Now, if the Beam target interlock shows GOOD, the gun is ready to accept triggers.

3.1.10 Phase shifters check

Phase shifters are used to adjust the RF signal phases in the accelerating sections or prebuncher cavity relative to a reference RF signal (generally relative to section 0). The phases must be checked to see that they can be adjusted.

- Operation of the phase shifters can be accessed on the Linux computers operating the general control interface, selecting the **RunLINAC** icon.
- Selecting the **Prebuncher** or any **Section #** button brings up a control screen where the RF **phase** of the Linac accelerating section or prebuncher can be controlled. Note: The section 1 phase shifter is also known as the High Power phase shifter.
- For each phase shifter, change the original number (remember it) in the edit box of the phase setpoint. If the feedback number also changed, this phase shifter is operational.
- Reenter the original number.
- If the feedback number is very far away from the setting number, or is unstable, the phase shifter needs to be serviced by the proper personnel.

3.1.11 Attenuators check

Attenuators are used to adjust the RF amplitude in the accelerating sections or prebuncher. They must be checked to see that they can be adjusted.

- Operation of the attenuators can be accessed on the Linux computers operating the general computer interface, selecting the **RunLINAC** icon.
- Selecting the **Prebuncher** or any **Section #** button brings up a control screen where the RF **attenuation** can be controlled.
- For each attenuator, change the original number (remember it) in the edit box of the attenuation setpoint. If the feedback number also changed, this attenuator is operational.
- Reenter the original number.
- If the feedback number is very far away from the setting number, or is unstable, the phase shifter needs to be serviced by the proper personnel.

3.1.12 RF drive setup

The RF drive (RF source) is located in rack R1032.2-01 of the modulator room. The RF drive houses a 2856 MHz crystal oscillator and a small klystron used to amplify the RF signal to roughly 8 kW which is fed to the inputs of six 20 MW klystrons. To prepare for optimization of the transmitters the RF START trigger is selected. Set up the RF driver, as follows:

- Initially, the RF drive must have the first three indicator lights lit (AC IN, AIR, CAB OVER/TEMP).
- Depress the **AC** push button located on panel P1032.2-02. This will supply 120V AC power to the cabinet and light the AC/ON indicator light.
- If it has not already been done, switch the **FIL AND 24VDC** and **RF** toggle switch up. This will turn the klystron’s filament on.
- After waiting about 5 minutes, the filaments will have warmed up and additional indicator lights will come on. The FIL, TRIG, DRIVER, and DOORS lights will light. The HV/OL and HV ENABLE lights may come on at this time.
- Press **HV/0VRESET** button. This will light the HV/OL and HV ENABLE indicator lights if they have not already been lit.
- Turn the klystron high voltage on, using the **HIGH VOLTAGE** POWER switch (yellow rocker switch) of PS1032.2-01.
- Press the green **HIGH VOLTAGE ON** button on PS1032.2-01. The RF drive high voltage should be set to 15 kV. This is displayed on an analog meter on panel P1032.2-01 and is adjusted using the knob labeled “Kilovolts”.

- Turn the **RF** key to **ON** position on P1032.2-02.
- If the RF driver works and the RF interlock is OK, the 8 kW RF power is sent out. One indication that the drive is operating is a frequency counter that is mounted in panel P1021.1-15 on the main console in the control room. A frequency very near 2856.000MHz should be displayed on the frequency counter.
- If there is still no RF signal shown on the frequency counter, check interlocks following the procedure outlined in 3.1.2 and the transmitters following the procedure in section 3.1.7.

3.1.13 Electron Gun Checks

The electron gun must be in proper operating order.

- The electron gun tank, located in zone 1 of the sub-basement, must be pressurized with 50 +/- 1 psi of SF6 to allow for adequate voltage isolation. As well, the gun cable must have 10 +/- 1 psi of insolating oil. These pressures should be checked if the gun has not been run for many days or weeks. If the pressure varies from these values, consult a qualified technical person to investigate.

3.2 Accelerating section conditioning using RF power

All accelerating sections in the Linac tunnel have to be conditioned with high power RF whenever the vacuum has been opened to the atmosphere or to nitrogen or when the linac has not been run for a long time (many weeks to months). This is because the gas, which is deposited on the section's wall, will be emitted when RF power heats the wall. This can result in large sparks. These sparks could lead to an avalanche condition which turns the vacuum extremely bad. The avalanche sparking makes shorts in the accelerating section and changes the impedance, resulting in large amounts of reflected RF power. This may destroy the klystrons. So it is necessary to get rid of the gas deposit on the wall by gradually feeding RF power into the sections--so called RF power conditioning.

For conditioning the accelerating sections, first complete the checks described in section 3.1 of this report, including the checks dealing with the vacuum, interlocks, cooling H/E, timing, RF drive, computer controllers and transmitters. When all these equipment is ready, RF conditioning can begin.

3.2.1 RF drive power setup

The RF drive is the common RF source for each high power klystron. A low power RF signal at a frequency of 2856 MHz from the RF drive is split seven ways and fed to six klystrons for amplification (with the seventh feed going to a test stand). After splitting, the amplitude of the RF power fed into each klystron can be adjusted by an attenuator. After the RF driver is turned on:

- Connect the RF drive signal from the **RF SOURCE** BNC bulkhead connector located on panel P1021.1-29 of the main control room console to an oscilloscope. If the RF drive has been turned on, the waveform will be displayed.
- If there is no signal, check if the driver is on (refer to section 3.1.12), and check the interlock. If there is still no RF, check the timing and follow the procedure outlined in section 3.2.2. The RF drive timing can be changed in the area labeled LINAC RF. This delay should be set to a value around 4370 ns.
- The duration of the RF drive signal is about 5 μs.
- The repetition rate of the RF drive can be from a minimum of a 0 Hz to a maximum of 2 Hz. (Above 2 Hz the RF power delivered to the accelerator sections is reduced and section design goals cannot be met.) Normally, it is set to 1 Hz.

3.2.2 Timing setup

The pulse duration of the RF is about 5 μs and the transmitter high voltage pulse duration is about 2 μs (the flat top is only 1-1.5μs). The timing relations between each transmitter and RF driver should be adjusted. To adjust the high voltage to coincide with the RF drive pulse complete the following procedure:

- Connect the transmitter's RF power **Fwd.** signal (located on panel P1021.1-12) to an oscilloscope.
- Switch to the desired transmitter using the 7 position rotary switch to adjust and monitor the selected RF waveform.
- To access the timing, pull down menu under the CLS logo and select **Timing** on the Linux computer interface. Make sure that the trigger mode LINAC/LTB is selected. This screen allows access to all of the delays associated with the modulators and gun operation.
- Do not change the number in the **LINAC RF** edit box.
- Type a new number into the modulator timing edit box, corresponding to the signal which has been connected to the oscilloscope. The waveform should be moving horizontally.
- When both leading and fall edges of the waveform are sharp, the bottom of the waveform is flat and the flat bottom is adjusted as wide as possible (at least 1 μs), the modulator's timing adjustment is done.
- Complete the two previous steps for all transmitters.
- The delay times can only be changed under the Machine Studies system mode, which can be accessed via the **CLS LOGO => Machine Setup**.

3.2.3 RF power conditioning and peaking

To run the accelerator sections at high power, remove the gas on the accelerating section's wall by gradually increasing the RF power fed into the sections. If sparking happens, a fault will occur that will kick off the RF (either VSWR or vacuum faults). The RF must be restarted. If many faults are occurring in rapid succession, the RF power may have to be decreased to continue the process. This process of increasing the RF power gradually to reach a maximum is called conditioning.

In general, an increase in the input RF power will result in the Klystron's output RF power increasing until the saturated point is reached. If the input RF drive power is increased further, the output RF power will decrease. Klystrons should be setup to operate right at the saturation point (peaked) for overcoming output RF power instability caused by input power fluctuation.

After modulator high voltages are set, all klystrons need to be "peaked". The input RF power to each section should be adjusted by changing the section attenuators.

- Connect the transmitter's RF power **Fwd.** signal (located on panel P1021.1-12) to an oscilloscope.
- Switch to the desired transmitter using the 7 position rotary switch to adjust and monitor the selected RF waveform.
- Operation of the attenuators can be accessed via the Linux computers operating the general computer interface, selecting the **RunLINAC** icon.
- Selecting any **Section #** button brings up a control screen where the **attenuation** leading to the klystron or the Linac accelerating section (in the case of section 1 or section 7) can be controlled.
- All sections may be conditioned at the same time.
- With a starting point of 15 to 20 dB, gradually decrease the number in the attenuation set point edit box by 0.5 dB steps (one step every 5 to 10 minutes is a reasonable rate).
- If sparking happens, a VSWR or vacuum "burp" could trip the RF. The RF can be reset by pushing the red **Linac Interlock Reset** button on panel P1021.1-22 of the main control console. If this results in an immediate trip, the attenuation number should then be set higher by 2 to 5 dB on the appropriate section, and the process can begin again from this point. Note: all trips should be recorded on log sheets.
- Continue monitoring the RF power during conditioning. When the RF power closes in on the saturation value, the step may change to 0.2 dB. Once the output power fails to increase for an increase to the drive power, or the output power is observed to decrease on the oscilloscope, the section is peaked. Set the attenuation to the value that is midway between where the decreases can be detected.

- **Note**: Some sections (currently section 6) may be experiencing some vacuum problem or larger than normal amounts of inside contamination. This may make it very hard to peak the RF power. In this situation, we can decrease the high voltage, thus decreasing the output power level at which it will be "peaked".

3.3 General adjustment and usage of elements

In general, operating the linac involves the control of many elements such as attenuators, phase shifters, magnets and all of the associated power supplies etc. The detail usage methods are described below.

3.3.1 Attenuation adjustment

The section attenuators for section 0, 2, 3, 4, 5, and 6 are used to adjust the RF drive power to the klystrons in order to control the RF power to the linac accelerating sections. The attenuators for the prebuncher and sections 1 and 7 are used to attenuate the high power RF reaching the equipment.

- Follow the procedure in 3.1.11 to access the control screen.
- Vary the number in the edit box to adjust the attenuation.
- **After the RF power is peaked, DO NOT CHANGE THE ATTENUATION AGAIN.**

3.3.2 Phase adjustment

The phase shifter is used to change the RF signal's phase between the accelerating section and a reference RF signal.

- Follow the procedure in 3.1.10 to get the control screen.

Three ways can be used to adjust the phase shifter. One is

- Directly put a degree number into the **phase set-point** edit box.

Second way is

- Follow the procedure described later in this report in section 3.3.9. Point the mouse to the number in the **phase set-point** edit box, then press, hold, and move it onto the knob manager.
- Turn the knob to change the phase.

Third one is

- After the phase control has been sent to the knob manager, right mouse select on the word **degree** > **SET** in the knob control window.
- A slider will appear that can be used to control the appropriate phase shifter**.**
- Similarly, the mouse arrow can point to **gain** and the adjustment gain can be changed.

- **NOTE: DO NOT USE THE *diagnostic* SCREEN TO CHANGE PHASE because this value will not be saved if it is accessed in this way.**

3.3.3 Magnet power supplies

Setup of the linac requires the adjustment of many different types of magnetic elements. Dipole magnets, quadrupole magnets, steering magnets, and solenoid magnets will all have to be controlled. The power supplies can be accessed via the **magnets** icon by pressing the **power supplies** button to reveal the **Linac and LTB Power Supplies** menu. All magnet power supplies servicing the linac should be checked to see that they turn on and enter the run state prior to setup of the linac.

For further details on the setting of magnets and general beam tuning concepts, refer to section 3 of the “Linac-to-Booster Setup Procedure” 2.9.36.1, R. Mark Silzer, 2003-02-07 (reference 8).

3.3.4 Dipole magnet adjustment

Dipole magnets are used to bend the beam through large angles in the Energy Compression System (ECS). The ECS dipoles should be degaussed when the beam is to proceed straight through the dipoles and cycled before it is set to a final value. Cycling allows the field to be reproduced with a greater accuracy. This is very important in reestablishing a setup after a power outage or some other interruption of power to the dipoles.

- Operation of the dipoles can be accessed on the Linux computers operating the general computer interface, selecting the **Linac & LTB Map** icon. The dipoles can be accessed from the resulting graphical interface.
- Select the **ECS to Bend** graphical interface.
- Select **B####-##** button to choose a magnet to be adjusted. This will initiate another control window for adjusting the selected dipole.
- The **Degauss** button can be selected to automatically degauss the dipole.
- The dipole must be set at zero after degauss. If not, set it to zero.
- To operate the dipole, it must be in **Normal** status. If not, set it to Normal.
- Cycling a dipole involves running the magnet at zero, waiting, setting the magnet to maximum value, waiting, and then setting the magnet to near its final value. The details of the cycling times are specific to each dipole.
- Adjustments to the dipole can be done without cycling by using the slider, knob box, or edit box. Ultimately, the dipoles should be cycled

to their final setting (although some very minor adjustment may still be required for fine tuning the field).

3.3.5 Quadrupole magnet adjustment

Quadrupole magnets are used to focus the beam. Operational experience has shown that Quadrupoles magnets do not require cycling. In a quadrupole magnet, the field reaches zero at the center of the magnet. It is convenient to use this property to define the magnetic center of the transfer line. It is advantageous to have the “steering out” of a quadrupole that is to be used to focus the beam. Thus, the quad can be used to adjust the size of the beam at a downstream location without changing the steering through the line. To take the steering out of a quadrupole,

- Operation of the quadrupoles can be accessed on the Linux computers operating the general computer interface, selecting the **Linac & LTB Map** icon. The quadrupoles can be accessed from the resulting graphical interface.
- **QF001-01&2** or **QD0001-01** are the quadrupoles that require tuning in the linac. These quadrupoles can be accessed via the **LINAC** graphical interface. Selecting the appropriate magnet name will initiate another control window for adjusting the quadrupole.
- The **Oscillate** button is used to vary the quadrupole field. The nature of the oscillation can be adjusted using the **Range** and **Period** edit boxes.
- The operator will view the beam on a downstream target while the quadrupole is set to oscillate around its running point.
- In general a “F-quad” (focuses in the horizontal plane) will generally have a larger effect in the horizontal plane, and a “D-quad” will have a larger effect in the vertical plane. However, QF0001-01, QF0001-02 and QD0001-01 are oriented as skew quads, so the deflections will be diagonal.
- With the quadrupole oscillating, the operator will adjust the upstream steering magnet to minimize the transverse kick given to the beam and position it at the center of the target location.
- If inadequate steering is available, it may be difficult to eliminate the quadrupole steering and keep the beam on axis. A tradeoff is often required.
- If the steering and positioning is very far off, this often indicates a problem with the upstream steering or that another magnetic element in the area of adjustment is powered or adjusted incorrectly.

3.3.6 Steering magnet and solenoid magnet adjustment

Steering magnets are used to make minor angular changes to the beam. Solenoids are used for focusing. Steering and solenoid magnets do not require cycling. Each accelerating section is equipped with a pair of X and Y steering magnets, used to effectively control the beam position and angle through the acceleration sections.

- Operation of the steering magnets or solenoids can be accessed on the Linux computers operating the general computer interface, by selecting on the **Linac & LTB Map** icon.
- Select the individual elements labelled **STxxxx-xx:X** or **STxxxx-xx:Y** or **SOLxxxx-xx** will initiate another control window for adjusting the selected element.
- The excitation to the chosen element can be adjusted by using the sliders, knob box, or edit box.
- When adjusting the magnet excitations, the beam position should be monitored on an appropriate pop-up monitor.

3.3.7 Slits adjustment

Slits are used to control the beam current and, in some cases, to limit the energy spread of the beam.

- Following the steps in 3.1.8, the operator can access the slits control window.
- The edit boxes can be used to open or close the gap between the slits, or move the center position. When adjustments are made, it is good practice to monitor the actual (feedback) number to verify the operation.
- If the actual number does not change, inform the proper technical personnel to solve the problem.

3.3.8 Pop-up monitors and view screens

The pop-up monitors along the linac and the view screens on the beam dumps or on the outrun window before the dumps act as visual indications of the position and shape of the electron beam.

- The pop-up monitors and view screens can be selected using the white pushbuttons on panel P1021.1-26 of the main control room console.
- The view screens can be viewed on the monitor screens mounted in the main control room console.
- The operator can use the view screens to see the beam position and shape.

3.3.9 Toroid, beam dump and FCT current monitors

The beam current signals from the toroids, beam dumps or fast current transformer (FCT) are the indicators of the beam current signal displayed in the time domain at those locations.

- The section toroid signals can be accessed via the BNC bulkhead connectors located on panel P1021.1-29 of the main control console. There are five connectors labeled **DC TOR**, **sec 0 TOR**, **sec 1 TOR, sec 2 TOR** and **sec 6 TOR** used to monitoring the beam current at these locations.
- There are two beam dumps that are used for monitoring the beam current in the ECS. The BNC bulkhead connectors labeled **ECS PIG 1** and **ECS PIG 2** for these dumps are located on panel P1021.1-19.
- All beam dumps are to be terminated into 50 Ω and have the calibration 20 mA / V.
- There is only one FCT current monitor. The N-type bulkhead connector labeled **FCT0003-01** is located on panel P1021.1-27. The FCT is to be terminated into 50 Ω and has a calibration of 0.4 A / V.
- Connect the toroid, beam dump or FCT signal from one of the bulkhead connectors to an oscilloscope terminated into 50 ohms (the exception is the DC toroid which is terminated into high impedance. If the electron beam is passing through the toroid or FCT or hitting the beam dump, the beam current waveform is displayed on the oscilloscope.
- DC toroid has the calibration 1 A / V.
- Other toroids have a calibration 0.2 A / V.
- The FCT has a calibration of 0.4 A / V.

3.3.10 Knob manager setup

The knob manager and knob boxes are supplied as an option to adjust an element's set point. To operate the knobs:

- Select on the **Knob Manager** icon on the desktop of the Linux control screen. A knob connection window will pop up.
- Place the mouse's pointer into the edit box containing the number value of the element being adjusted.
- Press the middle button on the mouse and hold it. A black sign indicating the signal to be controlled will appear.
- Hold and move the black sign into one of the small knob mange windows and release the middle button.
- Acknowledge the operation by pressing the **OK** button in the resulting window. The element's control is now connected to the desired knob.

- Turning the knob can change the element set point.

3.3.11 Collimators

The collimators are used primarily to protect downstream components from damage due to scattered or poorly steered beam. The operator needs to watch the beam passing through the collimator because the beam might be lost at these points.

4.0 Linac setup

ATTENTION: Before any part of the linac is turned on, both machine protection and personal safety interlocks must be in place.

This setup includes running the electron gun, prebuncher, section 0 - 6, the associated beam diagnostics, and the first portion of the ECS to optimize the energy and energy spread of the beam.

We assume that all required preparations described in the previous sections of this report have been completed. Completion of these procedures has established that RF power is being fed into the accelerating sections and that all equipment is ready for use.

4.1 Electron gun and optics setup

Setup of the electron gun involves the elements located before the prebuncher, i.e., gun, video deflector, collimators, lens1, lens 2, lens 3, solenoids and steering magnets. The gun controls are hard-wired and labeled as they came from the manufacturer. The gun controls are located on panels P1021.1-03, P1021.1-09, and P1021.1-10 on the main control room console. The elements they control follow the new CLS naming and numbering convention. The correlation is as follows:

GUN (steering)	ST0001-01
BIAS	STH0001-01 and STV0001-01
DRIFT	ST0001-02
ACCEL	ST0001-03
LENS1	SOL0001-01
LENS2	SOL0001-02
LENS3	SOL0001-03
SOLENOID 1	SOL0001-04
SOLENOID 2	SOL0001-05
SOLENOID 3	SOL0001-06

4.1.1 Gun set-up

For nominal operation, the gun must supply 1.2×10^{11} electrons in a 136 ns pulse. This corresponds to a DC current of 140 mA over the pulse duration. The energy of the electrons from the electron gun is 220 keV. The gun pulse (DC toroid) duration is controlled by timing signals to both the GUN and the VIDEO deflectors. The pulse shape and amplitude is optimized by use of the steering controls (BIAS and GUN) and the focusing controls (LENS1 and LENS2).

4.1.1.1 Gun Vacuum check

The electron vacuum pressure is interlocked and must be low enough (~10^{-8} Torr) before the electron filament can be turned on. Otherwise, the gas could damage the hot button cathode. The readiness verification is as follows:

- The electron gun vacuum control screen can be accessed via the pull down menu under the CLS logo, under **Linac -> Vacuum Control**.
- Details of the gun vacuum status can be obtained in the **Details** section by selecting the **Linac** button.
- Ion pump IOP0001-01 relates to the status of the gun vacuum. The ion pump will status will appear in the details window as either GOOD or BAD.
- The gun vacuum status also can be shown on the ion pump power supply **PS1021.5-01** located behind the main console in control room in rack R1021.5-06.
- If the vacuum level goes worse than the trip level, the power supply will draw larger currents and the vacuum valves will close. If the vacuum is very bad, this can cause the pump power supply to turn off. To restart the pump, just toggle the switch on the power supply off, and then turn it on again.

4.1.1.2 Gun Filament setup

ATTENTION: Before gun filament is turned on, the gun vacuum must be OK.

Once the gun vacuum has been verified “good”, the gun filament can be turned on. (This assumes that the gun has been previously conditioned. If the gun has recently undergone a major maintenance procedure, consult a qualified technician and follow the appropriate gun conditioning procedures (reference 3). Instruction and Maintenance Manual”. One copy of this manual will always be kept in the control room.)

- The master control panel for the gun is located on panel P1021.1-09 and is labeled: “INJECTOR STATUS AND CONTROLS”. By pressing the **INJECTOR->CONTROL->ON** button, all electron gun elements are made controllable.
- The next step is to turn the gun filament on. This is accomplished by pressing the **FILS VOLTS** button located on panel P1021.1-03 labeled: “ELECTRON GUN AND PULSER. The **FILS VOLTS** button should now be illuminated to indicate that the operator is able to raise or lower the electron filament voltage. The numeric LED display on the panel is used to indicate the current value of the illuminated control element.
- Continuously pressing the **RAISE** button will increases filament voltage (using **LOWER** button decrease it). Watch the gun vacuum and keep ion pump current less than 150 μA. It is good practice to use a camera

to display the gun vacuum on the main console. If the current is seen to increase very rapidly, stop and wait until it decreases to less than 50 μA.

- The gun filament voltage should be set at 117.1V. At this level, the filament current should be 1.82A. This is accessed by depressing the **FILS AMPS** button (both the voltage and current values are filament transformer primary side values, not the real filament voltage and current).
- The filament voltage might drop down after about 20 minutes due to gradual heating of the filament. After it is stable, the filament should be set back to 117.1V/1.82A.

4.1.1.3 Gun pulser set-up

Gun pulser controls the pulsed voltage between the gun grid and the cathode. This voltage controls the amount of electrons emitted from the gun.

- Before the gun pulser is setup, the gun filament must be set up in advance (see previous section).
- To set the pulser high voltage, pressing the **PULSER KV** button located on panel P1021.1-03 labeled: "ELECTRON GUN AND PULSER." The **PULSER KV** button should now be illuminated to indicate that the operator is able to raise or lower the electron filament voltage.
- Continuously pressing the **RAISE** button will increases the pulser voltage (using the **LOWER** button will decrease it) to normal operation value between 0.66 to 0.75 kV. The pulser also is equipped with a fine adjustment knob (rotary potentiometer) located to the left side of panel P1021.1-11.
- The relationship between the pulser and the injected beam current can be found in "Gun Running Data". This binder can be found in the control room.

4.1.1.4 Gun high voltage set-up

Before turning on the gun high voltage, all interlocks must be made, and the gun filament and gun pulser must be set. The gun high voltage controls the DC voltage between the gun cathode and anode. This voltage accelerates the electrons emitted from the gun cathode. To set the gun high voltage, follow the steps below:

- Turn the **GUN KEY #2** located on the right side of panel P1021.1-09 to the **ON** position.
- Press the **INJECTOR -> HIGH VOLTAGE -> ON** button located on panel P1021.1-09, labeled "INJECTOR STATUS AND CONTROLS". This will illuminate the red **ON** button.

- Press the **GUN KV** button located on panel P1021.1-03, labeled "ELECTRON GUN AND PULSER". The **GUN KV** button should now be illuminated to indicate that the operator is able to raise or lower the gun high voltage. There are two control modes, auto and manual.
- To chose the manual mode, the operator presses the **GUN KV MAN** button located to the upper right side of panel P1021.1-03. The high voltage is increased by pressing the **RAISE** button provided that the **GUN KV** button is still illuminated (using **LOWER** button can decrease it). The gun high voltage is normally operated at 220kV.
- To select the AUTO mode, simply press the **GUN KV AUTO** button located to the upper right side of panel P1021.1-03. . Do not confuse this button with the other **GUN KV AUTO** button located low on panel P1021.1-03. This button is used to preset the gun kV target set point, normally 220 kV (refer to the notes below). In the auto mode the gun high voltage will automatically increase to its target set point. The gun current should be 0.85 mA at 220 kV.
- Normally, the gun high voltage will be operated in the auto mode, and the voltage will automatically increase to 220 kV. If some sparking occurs, the gun high voltage will turn off. If this happens, the gun high voltage should be run up more gradually by using the manual mode. This is usually only a concern in the times following a major gun maintenance and reconditioning.

Important notes:

- To preset the gun high voltage target setting, press the **GUN KV AUTO** button (located low on panel P1021.1-03). The LED display will display the high voltage target setting in the AUTO mode. Hold the **GUN KV AUTO** button, and use a screwdriver on the potentiometer located below the button to change the setting value. (Generally, the operator is not required to complete this procedure.)
- By pressing the **GUN KV O/V** button the high voltage over-voltage trip value can be displayed and changed. Hold the **GUN KV O/V** button, and use a screwdriver on the potentiometer located below the button to change the setting value. (Generally, the operator is not required to complete this procedure.)
- By pressing the **GUN MAN O/I** button the gun current trip value can be displayed and changed. This is used to protect the gun when either a sparking or a short has occurred. Hold the **GUN KV AUTO** button, and use a screwdriver on the potentiometer located below the button to change the setting value. (Generally, operator is not required to complete this procedure.)
- When any interlock key is removed, the gun high voltage will drop down automatically.

- **<u>Do NOT press the HIGH VOLTAGE->OFF RESET button. It drops down the crow bar, shorting the high voltage to ground in the gun high voltage oil tank. This will require a new setup of the gun.</u>**

4.1.1.5 Gun beam current start

The gun beam current can be setup as follows:

- Note: Before turning on the gun beam current, the gun filament must be heated and high voltage must be set.
- Note: the **GUN KEY #1** and **GUN KEY #2** must be on. The GUN KEY #1 is located on P1021.1-22 and the GUN KEY #2 key is located on panel P1021.1-09 of the main control room console, labeled "INJECTOR STATUS AND CONTROLS".
- Press the **Beam -> ON** button located on panel P1021.1-09, labeled "INJECTOR STATUS AND CONTROLS". The **ON** button should be illuminated and the electron beam should be ejected out from the gun.

(a) Press the red **VAC VALVE VSWR RESET** button located on panel P1021.1-22 of the main control room console.

- To see if there is beam current, the operator can watch the gun vacuum ion pump power supply (**PS1021.5-01)** current meter located behind the main console in control room in rack R1021.5-06. If there is beam current, the ion pump will draw more current. Also, the beam can be seen either on an oscilloscope connected to the **DC** toroid (P1021.1-29) or on the first pop-up monitor labeled **END SEC 1** (accessed via P1021.1-26). If there is no RF in section 0 and section 1, you might not see it on the pop-up.
- If there is no beam, first check the **Machine Setup** and **Vacuum Control** as described in 3.1.9. and 3.1.1. respectively. Then the gun high voltage should be checked to be set at -220 kV, and the Micro84 programmable controller should be checked that all interlocks are ready for gun operation. If the programmer is functional, the green **Programmer OK** light on panel P1021.1-22 should be lit. On occasion the **Programmer OK** light is lit but there still is a problem with the controller. Cycling the **Programmer enable** toggle switch on panel P1021.1-22 down then up can sometimes reset the controller. If the **Programmer OK** light is not lit or is expected to be not functioning correctly, contact a qualified person to pursue a more thorough check.

4.1.2 Gun and video deflector timing adjustment

The electron gun timing signals includes **Gun On, Gun Off, Video Deflector On** and **Video Deflector Off**. Setting the gun timing is as follows:

(b) Follow the steps in section 3.1.4 to access the timing controls. The operation of the Gun and Video deflector timing can be checked by changing the numbers in the edit boxes and noting a change in the time of the beam leading and falling edges as viewed on the oscilloscope, externally triggered by the modulator triggers located in rack R1021.1-08 on F2N1021.1-02. Note: the delay times can only be changed under the Machine Studies or Maintenance system modes, which can be accessed via the **CLS LOGO => Machine Setup**.

- First, view the entire beam pulse without the influence of the video deflectors. To eliminate the effect of the video deflectors, push the blue **STEERING->BIAS** button located on panel P1021.1-10 (also known as ST0001-02) of the main control console and record the values of the bias steering. Turn the associated inner knob (horizontal) anticlockwise until a limit is reached. (Another way to remove the effect of the video deflectors is to move the video deflector triggers away from the gun pulse. This is done by moving the **VIDEO ON** towards 10 μs and **VIDEO OFF** >10 μs). The full gun beam waveform will be visible on the **DC** toroid (P1021.1-29) can be seen on the oscilloscope.
- Using the edit boxes in the timing window, set the **Gun On** timing to about 0 ns and the **Gun Off** to 5000 ns. The resulting beam width will be about 5 μs. At the same time, bring the transmitter 1 RF waveform (P1021.1-12) to another channel of the oscilloscope. If the beam is not in the RF duration, the Gun On and Gun Off timing will have to be readjusted.
- Gradually turn the inner **Steering->Bias** knob clockwise, until the video deflectors are seen to cut into the beam at the front and back of the pulse, and the beam pulse height is near a maximum. (If necessary, bring the video deflector triggers back using the timing edit boxes.) If the video deflectors are not effective in cutting the beam, the beam may be steered around the deflectors. This can be corrected by adjustments to the **GUN** (ST0001-01X&Y), **BIAS** (ST0001-02X & ST0001-03Y), **LENS 1** (ST0001-04X&Y) and **LENS 2** (ST0001-05X&Y), all located on panel P1021.1-10. If the video deflectors are still not effective, notify a qualified technical person to investigate.
- Adjust the **Video Deflector On** and **Video Deflector Off** timing to make the beam waveform top as square as possible. Generally, the VIDEO ON trigger is cut in to eliminate about the first 500 ns of the beam in order to obtain a very sharp leading edge to the pulse. The VIDEO OFF trigger is then cut in to obtain a 136 ns pulse.
- The gun and video deflector timing must next be adjusted together to ensure the beam passes through the accelerating sections when the RF power is present. Display the **DC** toroid signal (P1021.1-29) and the beam transmitter 1-6 RF forward signal (P1021.1-12) to the oscilloscope. By selecting the **PV GROUP** icon the **GUN DELAYS** slider can be used to move the beam pulse so that it is centered in the

RF pulse. (The **GUN DELAYS** moves the **Gun On, Gun Off, Video Deflector On** and **Video Deflector Off** delays all at once.)

4.1.3 Lens 1&2, Gun & Bias Steering adjustments

The electron beam pulse shape and amplitude is optimized by use of the steering controls (**GUN** and **BIAS**) and the focusing controls (**LENS1** and **LENS2**) located on P1021.1-10. Gun X&Y and BIAS X&Y steering magnets center the beam at the first and second collimators respectively. Lens1 and Lens 2 are used to focus the beam at the collimators.

- Adjust **GUN** steering, **LENS 1**, **BIAS** steering and **LENS 2** to maximize the beam current while maintaining a well defined 136 nsec pulse at the **DC** toroid as displayed on the oscilloscope. These controls require a substantial amount of adjustment in order to obtain a good beam.
- Finally, the gun pulser may need to be readjusted to give a DC beam current of 140 mA using the calibration: 1 A = 1 V into 1MΩ.

4.2 Prebuncher and Section 0 set-up

The prebuncher and Section 0 are used to bunch the beam into bunches with phase duration suitable for efficient acceleration through the remaining sections. Section 0 also has the function of accelerating the electrons to an energy of about 13 MeV. During the bunching and acceleration process, about one third of the beam is lost, leaving $6.8\text{x}10^{10}$ electrons in the pulse train (136 nsec). This corresponds to an average current of 80 mA in the pulse train.

4.2.1 The transmitter 1 RF power and timing set-up

4.2.1.1 Transmitter 1 RF power distribution

The prebuncher and Section 0 are supplied with RF power totaling about 13 MW. This corresponds to about two thirds of the 20 MW supplied from klystron 1. The RF power is distributed in the following way.

- First about one third of the RF power from klystron 1 is split off to be used in section 7, which is part of the ECS.
- The remaining two thirds of the power is used to feed the prebuncher and sections 0 and 1 where it is absorbed by an RF load. Thus, section 0 and section 1 are set up together since the RF loading can only be observed at the end of section 1. Most of the remaining two thirds of the RF power is used to power the two sections.
- A very small part (coupler –33dB) of this RF power is split off to the prebuncher. RF power levels and relative phase in the prebuncher are be controlled by an attenuator and phase shifter.

- Section 1 mainly serves to accelerate the electron beam. Under optimum conditions the beam energy after section 1 is about 43 MeV.

4.2.1.2 Transmitter 1 timing set-up

- Following the steps in 3.1.4, access the timing control window.
- Display the beam pulse (**Sec 1 Tor** P1021.1-27) and transmitter 1 **Fwd** (P1021.1-12) RF power waveforms on an oscilloscope.
- Adjusting the transmitter 1 timing, move the RF signal duration to cover the beam.

4.2.2 Set-up for steering magnets and solenoids

Before the prebuncher set-up, the beam must be brought to the pop-up beam monitor located at the end of section 1.

- The stepper motor driven monitor at the end of section 1 can be viewed by pressing the **END SEC 1** button located on panel P1021.1-26 of the main control console. The view screen is driven in by selecting on the **VSC0001-01** button, located on the Linac control map. This brings up another window that allows access to the viewer set point.
- Adjust **BIAS steering (ST0001-02X and ST0001-03Y)**, **DRIFT (ST0001-04X&Y)**, **ACCEL** (**ST0001-05X&Y)** knobs on control panel P1021.1-10 and **ST0001-06X&Y** on the Linac control map to make the beam centered at **VSC0001-01** while at the same time maximizing the beam current indicated by **SEC 0 TOR** (P1021.1-27).
- Adjust **LENS 3** (**SOL0001-03**, control panel P1021.1-10) and **SOL 1** (**SOL0001-04**; feedback display is on control panel P1021.1-10, adjustments are done via P1021.1-04) knobs to obtain a well-focused beam at **VSC0001-01** while maintaining the best beam pulse at the **SEC 0 TOR.**

  NOTE: Solenoids must be run at less than a value of 40 to avoid overheating of the coils.
- **SOL 2** (**SOL0001-05**; feedback P1021.1-10, adjustments P1021.1-04) and **SOL 3** (**SOL0001-06**; feedback P1021.1-10, adjustments P1021.1-04) are generally set to zero.
- If the prebuncher has been set correctly, the **SEC 0 TOR** should have about 80 mA of beam for the total duration of the pulse.
- These elements may require an amount of adjustment back and forth in order to obtain a good beam.
- After the prebuncher is set, it may be necessary to re-adjust these elements because the electromagnetic fields in the prebuncher and section 0 affect the particles' transverse movements.

4.2.3 Set-up for the prebuncher and section 0

The prebuncher is used to compress evenly distributed electrons in the time domain from the gun into about 60 degree of phase at 2856 MHz. The buncher or section 0 is used to compress the beam further to about 12 degree, and accelerating the beam to 13 MeV.

- The amount of RF power delivered to the prebuncher is set by adjusting the prebuncher attenuator (ATN0001-01). The prebuncher attenuator and phase shifter control can be accessed on the general Linux operating interface. Select the **Run Linac** icon, and then select the **Prebuncher** button to bring up a control screen in which the prebuncher **attenuation** and **phase** can be controlled.
- Bring the control onto the knob management following the steps outlined in section 3.3.10.
- Adjust the knobs to change the prebuncher's phase and attenuation. At the same time, the beam current is measured at **SEC 0 TOR**. The ratio of the currents measured at **SEC 0 TOR** and the **DC TOR** should be I_{SEC0} / I_{DC} = 0.66 (92mA/140mA) theoretically. However, operational experience has shown that best final energy spread results in only 57% of the beam (80mA) captured. (This step is only a rough adjustment to the prebuncher's phase and attenuation. Later on, they need to be readjusted to achieve the best energy spread at the ECS.)
- The **SEC 0 TOR** current can be optimized by using the gun focusing and steering controls: SOL0001-03 to SOL0001-06 and ST0001-03 to ST0001-06.
- Section 0 is supposed to be as an RF phase reference relative to all other sections, so it is not necessary to adjust its phase. Also, its RF power level does not need to be adjusted. (It has been set as outlined in section 3.2.3. Actually, the RF power from transmitter 1 has to be 20 MW so that the RF power at the beginning of section 0 should be about 12 MW.)

4.3 Section 1 set-up

Assuming the RF power amplitude has been set-up as described in section 3.2, section 1 can be adjusted. The set-up has four steps: timing, RF phase, steering and focusing.

4.3.1 Timing set-up

The timing of transmitter 1 is adjusted as described earlier in 4.2.1.2.

4.3.2 RF phase adjustment

Once the timing is adjusted the phasing of the section 1 RF is adjusted to maximize the loading as seen on the load of section 1.

- Following the steps described in 3.3.2., bring the **Section 1** control window up.
- Connect section 1 RF **Load** signal on panel **P1021.1-12** to an oscilloscope.
- Make note of the Deflector OFF timing set point and add 250 ns to this number in the Deflector OFF edit box. This increases the beam pulse width from 136 ns to about 400 ns. This will increase the beam loading so as to make it easy to observe the beam load effect when adjusting the phase.
- Viewing the effect of the beam on the LOAD signal is very difficult, and usually requires the use of a math function on an oscilloscope to display the subtraction of the signal with beam from a stored signal obtained when the beam was off. An example of such a setup is given below:

  Oscilloscope setup to view the beam loading using the Tektronix TDS 3054 or the TDS 3034B:

  - Press **BEAM->OFF READY** button on panel **P1021.1-09** to stop the beam.
  - To save the reference load RF waveform without beam:
    a. Press **SAVE/RECALL** button on the oscilloscope;
    b. Press **Save Waveform Chx** button under the screen;
    c. Press **To Ref1** button beside the screen to save it in reference 1.
  - Press the **VAC VALVE VSWR RESET** button on **P1021.1-22**.
  - Press **BEAM->ON** button on panel **P1021.1-09** to restart the beam.
  - To set up the math function:
    d. Press **MATH** button on the oscilloscope,
    e. Press **Set 1st Source** to **Ch1** (suppose the RF load waveform is on the CH1)
    f. Press **Set Operator** to **–**
    g. Press **Set 2nd Source** to **Ref1**.
  - Now, the MATH waveform equals to Ch1-Ref1. This action will eliminate most of the noise coming from the modulators.
- Following the steps in 3.1.10, access the section 1 phase control screen.
- Adjust the **high power->phase**. This is accomplished by typing a number into the edit box to change the phase or by bring it onto the

knob manager (refer to 3.3.10) to change it while monitoring the beam loaded RF signal. The more correct the phase is, the higher the amplitude of the signal. In this way, the best phase can be achieved. This procedure must be done in conjunction with steering and focusing described below (section 4.3.3).

- NOTE: the phase adjustments outlined here in section 4.3.2 will cause some minor steering through the section. The procedures outlined in section 4.3.2 and 4.3.3 below should be treated as an iterative process. If phase adjustment is done, the steering should be checked, or if the steering is adjusted, the phase adjustment should be checked and adjusted if necessary.
- Set the Deflector OFF timing back to its original value, producing a 136 ns pulse.

4.3.3 Steering and focusing

- Use steering magnets **ST0001-6X&Y** and gun steering **ACCEL** and **DRIFT** to maximize the signal on **SEC 0 TOR (TCT0001-2)** and center the beam on the view screen **END SEC 1** (**VSC0001-1**).
- Use quadrupoles **QF0001-01**, **QD0001-01** and **QF0001-02** to focus the beam on **VSC0001-01** (**QF0001-01** and **QF0001-02** are controlled together.) And use the quadrupole “triplet” to maximize the signal at **SEC 0 TOR**.
- Try to obtain the smallest spot with the lowest quadrupole values. **QF0001-01** and **QF0001-02** should be about 4.7A and **QD0001-01** should be about 7 to 8 A (when the PULSER is set between 0.1 to 0.7 kV).
- At this stage, the signal on **SEC 0 TOR** (**TCT0001-02)** may be optimized by adjusting the gun timing and section 1 RF phase as described earlier in sections 4.1.2 and 4.3.2.
- Take the steering out of the quadrupoles. Refer to Appendix A for guidance on steering through quadrupole magnets. Oscillate the quadrupoles of the triplet to see if the beam is steering at **VSC0001-01** and remove the steering by using **ST0001-06X&Y**. After the steering is removed, the beam can be centered (again) by using **ST0001-07X&Y**.
- This should maximize the current measured on **SEC 1 TOR** (**TCT0001-03**, P1021.1-29). Once again check the steering by oscillating the quadrupole triplet and correcting the steering if necessary.
- Use solenoids **SOL 1, SOL 2, SOL 3** (**SOL0001-04**, **SOL0001-05** and **SOL0001-06**; feedback P1021.1-10, adjustments P1021.1-4) to further focus the beam and maximize the beam current. SOL 2 and SOL 3 are generally set very close to zero.

4.4 Sections 2, 3, 4, 5 and 6 set-up

The RF power to each section that will be used has to be adjusted as described earlier in section 3.2.3, 3.3.1, and 3.3.2. Proper acceleration in each section requires that each section is running at the correct temperature. The heat exchangers for each section must be operating and regulating the temperature as required. Sections that are not used for acceleration are detuned by changing the temperature of the section well away from the operating temperature (55°C).

Similar as the section 1 set-up, set section 2 to 6 as follows:

4.4.1 Timing set-up and jitter elimination

There could be time jitters among the transmitters. The jitter must be checked and reduced by adjusting the transmitter's timing while maintaining the RF pulses at an appropriate temporal location with respect to the beam pulse. These jitters could result in significant energy jitters in the electron beam as measured later at the ECS.

- Following the steps in 3.1.4 to access the timing control window.
- First the timing of sections 2 to 6 are pre-adjusted so that the front edge of all RF pulses starts at a similar time as the RF pulse from section 1 as viewed on the oscilloscope.
- On two different channels of an oscilloscope, display the RF pulse of section 1 and 2. It is convenient to use the forward pulse of one section and the load pulse of the other section (P1021.1-12). Alternately, the modulator Voltage pulse and Current pulse can be used to set the timings (P1021.1-13).
- Measure the timing jitter between the two pulses. The jitter should be between 2.4 and 7.6 ns. If any one is bigger than 8 ns, adjust the timing of section 2 to reduce the jitter. It is also important to check to see that adjusting the timing of one section does not move the timing of the other. This should be checked, and the timing should be adjusted to a position where no "cross-talk" between the sections is observed. If no place can be found within about 50 ns of adjustments where the sections are not seen to follow one another, report this to an appropriate member of the technical staff.
- Do the checks and adjustments listed in the step above for sections 3 and 4 and sections 5 and 6.
- As a final check, if the transmitter noise jitter is adjusted correctly, the noise jitter measured on the signal from the **SEC 1 TOR** (P1021.1-29) must be no more that 10 ns with respect to the MOD trigger (F2N1021.1-02).
- Note: Section 3 has been used to overcome the energy spread caused by beam loading in the sections. Thus, section 3's timing needs to be readjusted later.

4.4.2 Phase set-up

For proper phasing of each section, the RF power to each section must be controlled within one degree of phase at 2856 MHz (about 1 ps).

- Following the general procedure described in section 4.3.2, the phase to sections 2, 3, 4,and 5 can now be adjusted. Section 6 is kept for compensating beam energy to designated value (normally 250 MeV), and thus, it is not adjusted in the same manner.
- The phasing of the section RF is adjusted to maximize the loading as seen on the load of the section (RFL0001-02 to RFL0001-05). As with the adjustments to section 1, the phase adjustments must be done in conjunction with steering and focusing described below.

NOTE: The section loading, steering and focusing must be adjusted with great care and attention to detail. When this is done, the section phasing should be close enough to so that the resulting beam energy is very near the optimum energy obtainable from the sections. Subsequent phase adjustments required to fine tune the beam energy and energy spread should be small enough so that they will have only a minimal effect on the trajectory of the beam.

4.4.3 Steering and focusing

This procedure is similar to the one described in section 4.3.3. The following section is to be used as a guideline for adjusting the steering and focusing through the sections. There is no set sequence for this tuning, and it is an iterative process also incorporating the phase adjustments of the sections described above in section 4.4.2.

- When a section’s phase has been adjusted, this can result in transverse deflections of the beam. Thus, appropriate steering is use to center the beam on the related pop-up screen. For adjustments to sections 2, 3, and 4, the pop-up **END SEC 4** (P1021.1-26) is usually used. For sections 5 and 6, pop-ups **END SEC 5** and **END SEC 6** (P1021.1-26) are used. Operational experience has shown that it is not best to center the beam at these two view screens. At these two points, the beam is usually set to the left part of the screen. In the future we will try and have photos or diagrams located in the control room to identify the current preferred locations of the beam spot on the screens.
- Initially, the steering through the sections should be set to zero. The steering is adjusted sequentially to avoid too much ”dog legging” down the linac.
- If the beam is centered on a pop-up screen and all downstream steering is set to zero, but the beam spot at the downstream pop-up is far from the center, you should proceed as follows: Try to adjust the RF phase of the downstream section first and see if the steering

improves. If the phase of the downstream sections has been optimized, adjustments to the steering magnets associated with the section are used. As well, minor adjustments to steering magnets located further upstream can be done to minimize the current on any one steering magnet.

- The solenoids SOL0001-07, SOL0001-08 and SOL0001-09 are used to maintain a small beam size, conveniently viewed at the pop-up **END SEC 4** (P1021.1-26). The solenoids are accessed via the control system graphical interface.
- The quad pair QF0001-01 and QF0001-02 and the single quad QD0001-01 are adjusted to give a well-focused beam. This is viewed at the **END SEC 5** and **END SEC 6** pop-ups (P1021.1-26).
- Remember: The **START SEC 2** pop-up is located before section 2 (Not at end of section2). So adjustments to any element related to section 2 will not change the beam at this point.
- Adjust ST0001-06X&Y targeting **END SEC 1**; ST0001-07X&Y targeting **START SEC 2**; ST0001-08X&Y targeting **END SEC 3** or **END SEC 4**; and ST0001-09X&Y targeting **END SEC 4**.
- There is only one set of steering magnets between section 5 and 6. Adjust ST0001-10X&Y, targeting both **END SEC 5** and **END SEC 6**. This will require making tradeoffs between the positioning at the two viewers. (The ST0001-11X&Y are too weak to bend the beam adequately and are useful only at low beam energies).
- In some cases, tradeoffs must be made between the steering and the downstream position of the beam if the full complement of steering magnets is not available.
- The beam pulse should now be observed by use of the current monitor ICT/FCT10003-01 (FCT is accessed via P1021.1-27) located beyond the ECS (the ECS magnets need to be degaussed as described in section 3.3.4). The current observed at this current monitor should be about 95% of the current observed at **SEC 0 TOR** (P1021.1-29). Failing to transmit 95% of the current through the accelerating sections indicates that more steering, focusing and phasing adjustment is required.

NOTE: The procedure outlined from section 4.2. to 4.5. may need to be revisited many times to obtain a good beam. This is mainly due to the coupling that exists between longitudinal and transverse components of the beam.

4.5 Energy, Energy Spread set-up at ECS

Beam energy, energy spread and emittance are main beam parameters that can be adjusted and measured using the ECS. Correct energy setting is achieved by degaussing and then correctly setting the first dipole of the ECS. Subsequent adjustments to the accelerator sections are used to match the beam energy to the setting of this dipole. A tight energy spread is obtained by adjusting the phasing and RF amplitude in the prebuncher and accelerating sections. Also, energy spread is adjusted by using the timing of section 3 and section 4.

4.5.1 ECS first bending magnet and energy set-up

- The dipoles B0002-01 and B0002-02 must be degaussed.
- Using dump BST0002-01, set the first ECS dipole B0002-01 close to the approximate beam energy (the most current calibration will be implemented in the dipole control window), following the steps in section 3.3.4.
- The phase of section 6 can be used to fine tune the energy of the beam. Now adjust section 6 phase to obtain the beam on the BST0002-01 (signal is on P1021.1-19 labeled ECS Pig 1). The beam spot can be viewed on **ECS CAM 1** (P1021.1-26). In general, the beam can be centered on the viewer for two different phases. Note these phases and continue operating at the one with the higher phase angle, as this phase will help aid in the bunching of the beam.
- To determine the maximum energy of the linac, adjust the phase of each section to move the beam to the right as viewed on **ECS CAM 1** (P1021.1-26). As the beam moves right (higher energy), it can be brought back to the center of the screen and a stable pulse can be maintained on BST0002-01 by changing the section 6 phase.
- The phase adjustment required at this stage should be very small, and consequently will have only a minor effect on the steering. However, it is good practice to quickly check the steering down the linac and to make any small adjustments that are necessary (section 4.4.3).

4.5.2 Optimizing the Beam Energy Spread

Good energy spread can be achieved by adjust to the section phases, prebuncher phase, prebuncher attenuation and timing of sections 3 and 4 (group adjustment). These adjustments are iterative, and may have to be done a number of times to obtain the desired result.

4.5.2.1 Adjusting the prebuncher’s phase and attenuation

- Before these adjustments, all section RF phases must be optimized and B0002-01 set to the correct value (section 4.5.1).
- The beam must have also been brought on the middle of the viewer before the dump BST0002-01.
- Adjust the prebuncher phase and attenuation to get a bright beam spot on **ECS CAM 1** (P1021.1-26) with as small a horizontal spread as possible.

4.5.2.2 Adjusting the Timing of Section 3 and 4

- The section 3 timing control should be combined with the section 4 timing because it could cause time jitter, changing the time relation between section 3 and 4. Use the **Pvgroup** control screen which can be accessed via the **CLS LOGO =>PVGroup.**
- By selecting besides the slide button, the operator can change the section 3&4 timing--1ns per click.
- Change the section 3&4 timing gradually. At the same time, change section 6 phase to keep the beam spot on the middle of the viewer. If the energy spread gets worse, adjust the timing in the other direction.
- Continue to change the timing until the best beam energy spread is obtained.

4.5.2.3 Adjusting prebuncher and sections' phase again

- Now, adjust the prebuncher's attenuation and phase slightly, to optimize the energy spread.
- Adjust the section 1 to section 5 phases again. The same energy gain can be given to the beam at two different phases, one at a higher phase angle and one at a lower phase angle. In general you should note these phases and continue operating at the one with the higher phase angle, as this phase will help aid in the bunching of the beam. Adjust these phases to result in a beam with the lowest energy spread. The beam can be kept at the center of the viewer (**ECS CAM 1**) by using section 6 phase.
- A good energy spread in the beam is evident when the spot on the **ECS CAM 1** viewer at the outrun window of B0002-02 appears round, with very little horizontal spreading. The spot should only extend horizontally about one tenth of the way across this viewer. The energy spread can be checked more thoroughly by the method described in section 4.5.5.

4.5.2.4 Setup of ECS Slits CLH0002-02

- The control of the slits can be accessed via the **CLS LOGO => Linac => Run Linac** or via the **Run Linac** Icon and then selecting the **linac Slits** button.
- The center position of slits CLH0002-02 is very important in the transport of the "on-energy" electron beam through the ECS system. Set the center position of CLH0002-02 to the current setup position (consult previous setup data).
- Adjust section #6 phase to maximize the beam current on BST0002-01.

4.5.3 ECS Setup

Detailed setup of the ECS under normal operation is outlined in the Linac-to Booster Setup Procedure, 2.9.36.1 (reference 8). Preliminary setup of the beam straight through the ECS is required to ensure that the beam has been positioned properly for the detailed setup.

4.5.4 ECS Straight Through Operation (Option)

Theoretically, before setting the ECS, the beam should go straight through the ECS. However, operational experience has shown that it is better to steer the beam towards dipole B0002-02. This results in the beam being positioned on the left side of the view screens **END SEC 5** and **END SEC 6** (P1021.1-26). .

- Open the emittance slits, CLH0002-01, CLV0002-01 and the energy slits CLH0002-02 to 20 mm for setting up the ECS. The slits can be controlled via the **RunLINAC** icon and selecting the **Linac Slits** button.
- Before the ECS is used, the ECS magnets B0002-01, B0002-02 and B0002-03 must be degaussed so that the beam can be transported straight through. See the procedure in 3.3.4.
- Using ST0002-01X&Y and ST0003-01X&Y center the beam at TRM0003-01. (Note: TRM0003-01 is a phosphor view screen).
- Look at the FCT0003-01 signal (P1021.1-27) to aid in the initial steering. Now use the ST0003-01 steering magnets to maximize the current on the current monitor and to obtain a square pulse. To fine tune the steering, bring the emittance slits to 8 mm and use the steering magnets to maximize the beam current.
- If required, the beam can be focused with quadrupoles QD0003-01 and QF0003-01. These quads can be oscillated and ST0002-01X&Y and ST0003-01X&Y can be adjusted to minimize the quad steering while keeping the beam centered on TRM0003-01 (refer to Appendix A).

Proper set up of the straight through operation should allow the electron beam to proceed down the beam pipe beyond the ECS to the straight through dump. At this stage however, the beam is not focused for optimum dumping of the beam (BST0003-02) and further transport through the LTB line. To transport the beam through the LTB without using the ECS, or for an accurate measurement of the beam current at BST0003-02 the beam can be focused using the “phase space” quadrupoles, (QD0003-01, QF0003-01, QB0003-01 and QD0003-02). Operational experience has shown that a well focused beam is obtained by powering QF0003-01 and QD0003-02 and setting QD0003-01 and QB0003-01 to zero. For details of the transport to BST0003-02, refer to the Linac-to Booster Setup Procedure, 2.9.36.1 (Reference 8).

4.5.5 Measuring the Energy Spread Using the ECS

- The energy spread of the beam can be roughly measured by comparing the horizontal beam size at **ECS CAM 1** to the horizontal movement observed on the screen for different excitation values of the dipole B0002-01. As an example, if the beam can be scanned across the entire screen at the B0002-02 outrun window (**ECS CAM 1**) by an excitation change of 10 MeV as read from the B0002-01 calibrations, and the beam is seen to span one fourth of the screen, then the energy spread would be roughly 2.5 MeV. This would correspond to an energy spread of about 1 % at a beam energy of 250 MeV.
- The energy spread of the beam can be checked by observing the current on BST0002-01 for two CLH0002-02 slit settings. With the slit gap set at 7mm, the beam current signal should have greater than one half of the signal that is observed for a setting of 14 mm (Theoretically, half the full current with 7mm gap corresponds to a 1% energy spread). This can be optimized by using section 6 phase and may require adjusting steering magnet ST0002-01. (Operational experience has shown the signal at 14 mm should be close to 100% transmission and very close to full current transmission can be obtained at slit values as low as 8mm for a very good setup).

4.5.6 Emittance measurement

- See reference 7.

5.0 Reference

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7. The Linac-to-Booster Transfer Line Further Refinements 2.2.69.2, Jack Bergstrom, 2000-02-09.
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6.0 Appendix A, Steering through quadrupole magnets

In a quadrupole magnet, the field reaches a zero at the center of the magnet. It is convenient to use this property to define the magnetic center of the transfer line. If the beam goes through the center of a quadrupole, any change in the strength of the quadrupole field will not cause any steering of the beam, but a change in the focus will be observed. Conversely, if the beam is traversing the quadrupole off axis, a noticeable shift and focusing of the beam will be seen simultaneously on a downstream target when the quadrupole field is changed. It is advantageous to have the “steering out” of a quadrupole that is to be used to focus the beam. Thus, the quad can be used to adjust the size of the beam at a downstream location without changing the steering through the line. Some quadrupoles are used to establish specific optical properties in the line between two points in the lattice, and are to be run at close to the theoretical values. It is less important to have the steering out of these quadrupoles since they are to run at one field. However, having the steering out of these quadrupoles can be used to determine if the beam is going through the center of the transfer line (see section 3.3).

The steering is taken out of a quadrupole with steering coils located upstream from the quadrupole location (NOTE: Optimally, two sets of vertical and two sets of horizontal steering coils are used to fully define the angle and position through the quadrupole. An upstream dipole can be used in some locations where inadequate steering coils are available. In some locations, tradeoffs must be made between the steering and the downstream position of the beam if the full complement of steerers are not available.). A downstream view screen (TRM or other target) is used to view the beam location and shape.

To take the steering out of a quadrupole, the operator will view the beam on the downstream target while the quadrupole is set to oscillate around its running point. An “F-quad” (focus in the horizontal plane) will generally have a larger effect in the horizontal plane, and a “D-quad” will have a larger effect in the vertical plane. With the quadrupole oscillating, the operator will adjust the upstream steering to minimize the transverse kick to the beam and position it at the center of the target location. If inadequate steering is available, it may be difficult to eliminate the quadrupole steering and keep the beam on axis. A tradeoff is often required. If the steering and positioning is very far off, this often indicates that there is a problem with the upstream steering or that another magnetic element in the area of adjustment is powered or adjusted incorrectly.